

LESSON 5-8 • SESSION 2 • PRACTICE • FORM A**Lateral Area of Pyramids**

Learning target – I can find the lateral area of a pyramid, which covers only its slanted faces. • 6.G.A.4

REMEMBER

- **Lateral area** = the slanted faces only. Do **not** include the base.
- **Surface area** = lateral area + base area.
- A triangle-base pyramid has **3** slanted faces; a square pyramid has **4**.

WORKED EXAMPLE

Square base 4 m; slant height 6 m; only the sides are covered.

Step 1 Leave out the base.**Step 2** One triangle: $\frac{1}{2} \times 4 \times 6 = 12 \text{ m}^2$.**Step 3** Four triangles: $4 \times 12 = 48$.**Answer lateral area = 48 m²****WORDS** lateral area (*área lateral*) • surface area (*área superficial*) • slant height (*altura inclinada*)**PRACTICE**

show your work • label units

- 1.**
- Find each lateral area.

a. 4 faces, each 25 cm^2 ____ cm^2

b. SA = 150 in^2 , base = 36 in^2 ____ in^2

c. square base 7 m, slant 9 m ____ m^2

d. square base 11 in, slant 10 in ____ in^2

- 2. Canvas Frame.**
- Maria's frame is a pyramid with a triangle base (24 in sides) and slant height 14 in. Only the slanted faces get canvas.

a. Faces that get canvas _____

b. One slanted face _____ in^2

c. Canvas needed _____ in^2

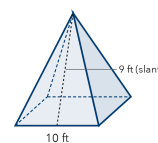
Show your work.

- 3. Open-Bottom Tent.**
- This tent is a square pyramid. The floor is open ground.

a. One panel _____ ft^2

b. Fabric for the tent _____ ft^2

c. Total if the floor were covered _____ ft^2



Show your work.

- 4. Gazebo Roof.**
- A square pyramid roof has base 14 ft and slant height 8 ft. Only its slanted faces get shingles.

Show your work.

Answer _____ ft^2

- 5. Tower Cap.**
- A square pyramid cap has base 22 m and slant height 17 m. Only its slanted faces are copper.

Show your work.

Answer _____ m^2

- 6. Display Pyramid.** A display is a square pyramid: base 26 in, slant height 25 in. Every face is wrapped, base included. How much paper?

Hint Include the base.

Show your work.

Answer _____ in²

- 7. Kai's Canvas.** A square pyramid has base 30 cm and slant height 21 cm. Kai says the four slanted faces cover 2,520 cm². Is Kai right?

Hint Find one slanted face first.

Show your work.

Answer _____ cm²

- 8. Tent Cover.** A shelter is a square pyramid: base 16 ft, slant height 10 ft, open floor. Canvas costs \$3 per ft².

Hint Find the lateral area first, then multiply by the price.

- a. Lateral area _____ ft²
 b. Cost of the canvas _____
 c. Total SA; why use lateral area here? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Rosa's Work.** A square pyramid has base 14 cm and slant height 8 cm. Only the slanted faces are painted. Rosa wrote: "Area = 196 + 224 = 420 cm²." Explain the mistake and fix it.

TEST PRACTICE

choose the best answer

- 10.** Lateral area leaves out the –
A base **B** triangles **C** slant height **D** apex
- 11.** How many slanted faces does a pyramid with a triangle base have?
A 3 **B** 4 **C** 5 **D** 6

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

2,520 Q	1,976 G	1,260 L	180 H	320 E	1,232 Z	224 I	960 S	504 S	748 N
#2c	#3b	#4	#5	#6	#7	#8a	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____